Financial Sentiment Across News Sources

PROJECT PROPOSAL

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Repository:
https://github.com/Roee104/NLP-financial-sentiment-analysis.git

Problem Statement

- **Motivation**

  Financial news plays a critical role in influencing market trends, and quick interpretation of this news can empower investors and analysts

- **Problem Definition**

  - *NLP Task:* Develop an LLM-based system that automatically classifies financial news articles into sentiment categories: Positive, Negative, or Neutral. (Optionally, provide sentiment intensity scores to gauge confidence)
  - *Input*: Text from financial news articles (headlines, snippets, or full articles).
  - *Output*: Sentiment labels with optional confidence scores.

- **Challenges**

  - Handling domain-specific language, subtle sentiment cues, and the unique financial jargon present in the news.

Training and Test Data

Type

- The datasets consist of financial news articles with sentiment annotations (Positive, Negative, Neutral)

Sources

- Utilize publicly available datasets like the Financial Phrase Bank and relevant financial news datasets on Kaggle.

Evaluation

Metrics:

- Accuracy, Precision, Recall, and F1-score to measure sentiment classification performance

Baseline:

- Candidate sentiment predictions are ranked by computing cosine similarity between FinBERT-derived text embeddings and entity embeddings from expert annotations.